Assignment Introduction

You are tasked with designing **a web-based communication system** to support university students doing **group work** (e.g., assignments and projects). Your design should go *beyond existing tools* like Slack, Discord, or Teams by focusing on the unique needs of students and their group work.

In this project, you will follow a *User-Centered Design (UCD)* process to **identify user requirements**, **design alternatives**, and **develop *lo-fi* prototypes**. Then, you will use a selected set of Large Language Models (LLMs) to assist in **developing *hi-fi* prototypes**.

Considering real world deployment, you will define **security requirements** for your system. You will **evaluate vulnerabilities** in the generated systems and apply **security solutions** to address them.

Finally, you will create an **evaluation plan** to see how they meet the design and user requirements.

Things to note:

- The project is divided into three (3) phases, and each phase has a specific assignment that you have to submit on Canvas. Phases are designed so that you can chronologically follow them with overlaps.
- This is a group project. There should be 4 students per group, and where an exception can be made, groups of 3 students are allowed. You must make a case to your tutor why it has to be a group of 3. All the group members must be from the same tutorial group.
- The groups will be formed in the W01 tutorial with the students present during the tutorial. Once you form the group, inform your tutor; they will set up your group in Canvas.
- We highly encourage you to use modern development tools (e.g., AI-assisted IDEs like Cursor or Windsuf) to minimise overhead, allowing you to focus on usability and security concepts.
- Keep all of your answers concise and on-point, and do not repeat yourself. We highly encourage you to develop the report concurrently while working on the project tasks, rather than starting the report after all the tasks are completed.
- The submission must be made in PDF format. All submitted *text* (report, codes, etc.) must be in *text* format that is readable by Turnitin. Do not convert the text part of the report to images (e.g., jpeg). If you submit in such a format, we may ask you to resubmit the report, and you may be subject to late submission penalties. Please read the **Adobe PDF** section of the Turnitin file requirements section to see how you can ensure it is readable. The images in your report (e.g., images of your prototype) do not need to meet this requirement. LaTeX generated PDF files naturally meet this requirement unless you further process them.
- Every project member should contribute equally to the project (both the project tasks and writing the report) and **individual contributions should be reflected using the diary section of the report (Page 5)**. All the project-related code should be managed through the USyd Github.
- **Report Template:** The template for the report will be provided in Canvas.
- **All the project members' names, SIDs, the project group name (as it appears in Canvas) should be included as the author information.** Incorrect template, missing author or group information will incur penalties (-10% per phase).

Phase 1 - Project Plan (50 pts [weight 5%])

In this phase, you are required to develop the project plan and the scope. This involves several tasks, as outlined below. Each task should be recorded in the report following the template. Please follow the word counts per task where they apply. Also, include the *diary* for the phase as outlined on Page 5 at the end of the report.

Tasks:

1. Formulate a preliminary system description for the proposed system. It must include:**- *Task total 9pts***
   (a) The goals and aims of the system (less than 50 words).
   (b) Ideate at least 1 feature you could integrate into the proposed system that current communication platforms do not have (less than 30 words).
   (c) State how the above feature will be helpful for students to do their group work (less than 30 words).
2. Selecting potential implementation mechanisms. **- *Task total 41pts***
   (a) Survey and list 6 (4 for 3 student groups) different generative AI platforms (with URLs to their website) you can use to help implement the system above. They should not be different versions of the same platform. You should not use *DeepSeek* as the university suggests you not to. We encouraged you to use freely available versions.
   (b) For each GenAI platform above, list one potential advantage and one disadvantage of using them to help implement the proposed system. This can be based on the known capabilities and issues of the selected platform.
   (c) Write a prompt that you imagine will help you to develop a simple prototype for the web interface with basic functionality for a quick test of the proposed system.
   (d) Using the same prompt above on each GenAI platform above, generate different prototypes for the proposed system. There should be 6 (4 for 3 student groups) prototypes generated per group. Try them out and add screenshots of the developed prototype system (1 to 2 screenshots per platform is sufficient) showing key features to your report.
   (e) Examine the design and the functionality of the generated interface, and rate each prototype generated by the GenAI platforms, and describe the reasoning for your rating (less than 50 words per platform).

Phase 2 - Requirement Gathering & Design (100pts [weight 10%])

In this phase, you will gather user requirements, develop conceptual designs and prototypes for your group project. This involves several tasks, as outlined below. Each task will either have a reporting or a live demo during the tutorial. Also, include the *diary* for the period as outlined on Page 5 at the end of the report.

Tasks:

1. Identify user requirements for the project. ***- Task total 40pts***
   (a) Develop a plan for a semi-structured interview to gather requirements for the proposed project. The plan must include: ***- 10pts***
   - The goal and a brief description of the interview.
   - Who the potential participants are, along with the inclusion/exclusion criteria.
   - Plans for pilot studies and ethical considerations (including required ethics documents)[1].
   - Planned activities, interview questions, data to be collected, data recording formats, and analysis plan.

   (b) Recruit participants and conduct the interview. Include transcripts in the appendix. Groups of 4 must interview 4 participants, where groups of 3 - 3 participants. Participants must be recruited outside the INFO2222 (2026) cohort. ***- 10pts***

   (c) Conduct a thematic analysis. Document the process followed and the results (themes). ***- 10pts***

   (d) Using the themes as references, create a table of features required for your new system along with a short justification for each. Clearly state if the selected features are the same, refined from or new compared to the features you came up with in Phase 1. Compare these to existing systems (Teams, Slack, etc.), and clearly outline if each feature exists in them. We recommend you find at least 1-2 unique features matching the context of the project scope. ***- 5pts***

   (e) Write a *user scenario* of how your proposed project will be used. Please read these articles to learn about how to write a *user scenario*: [article 1, article 2] ***- 5pts***

2. Designing alternatives. ***- Task total 20pts***
   (a) Follow the "10 plus 10 method" to sketch your ideas for a potential web-based user interface to implement the system. As a group, select 2 alternative designs. Please briefly explain (max 200 words per design) your two ideas in the report (include your sketches from the selected designs, limited to 6 sketches per design). ***- 10pts***

   (b) Either choose between (and/or) merge the two designs into one design solution and justify it with your reasoning. Develop a paper prototype (vertical + horizontal) for your chosen design. For the vertical prototype, we recommend you focus on one key feature. Include photos of your paper prototype in the report. Paper prototypes should be hand-drawn (you may use a digital device to draw if preferred), and you must have physical paper prototypes (e.g., printed) to be shown during the tutorial. ***- 10pts***

3. Hi-Fi prototypes. ***- Task total 10pts***
   (a) Using either an AI tool you explored and found to be useful in Phase 1, or using a prototyping tool like Figma, Stitch, etc., develop a high-fidelity prototype. The following considerations should be taken into account: ***- 10pts***

[1] Please check the HREC guidelines of the university, for this study, you will not need the ethics approval since we will not publish the results. But you should download the template for consent and information sheets, and include those into your study process, i.e., collect informed consent.

- Use the paper prototype as the basis for your hi-fi prototype. You can use the aspects of it to generate prompts, or even directly use them as input screenshots to generate the hi-fi prototype.
- We recommend following technologies (you can integrate those into your prompts), but you don't have to limit yourself for these. Implementation choices are yours. These are very popular technologies that may have a lot of resources available online. And also, if you use these technologies, we will be able to help you with some technical queries along the way.
  - Next.js: A framework for both the frontend (UI) and the backend server logic. Best to use `App Router` and `TypeScript`.
  - Tailwind CSS: An easy to use CSS framework to design responsive web interfaces.
  - Socket.io: A powerful JavaScript library for real-time, bidirectional, event-based communication (for the chat). You can use it for features like instant message broadcasting, online/offline status, and notifications.
  - You may also consider using a simple local database, e.g., `SQLite`, Local `MongoDB`, or `RxDB`.
  - While the above are recommendations, you may choose the appropriate technologies based on experience and knowledge of the team members. For instance, if you are familiar with Python, technologies like `Flask` or `Django` may work better.
- You must clearly document all prompts and AI-generated content produced using AI-assisted tools as an appendix in your report, for reproducibility and to address plagiarism concerns. Any file, function, or substantial code snippet created or modified by AI assistance must be explicitly listed (see Table 1 for an example).
- Please provide your source files (USyd Github link in the report) in your final submission with documentation (in a `README.md`).
- Make sure the system has some functionality (it is OK if not all the functions are working) at the live demo during the tutorial (Week 06). You must clearly indicate which functions do not work in the report.

(b) Live demo during tutorial, showcase your themes, lo-fi and hi-fi prototypes (5 mins per group, followed by 5 mins Q&A). The attendance during the demo and Q&A is needed to gain the points for this part of the assessment. ***- Task total 30pts***

File Path	**Description**	**AI Tool**	**Prompt Used**
`hello.py`, `greeting.py`	Scripts printing greeting messages	claude-3.7-sonnet	"Write Python scripts that greet the user differently based on the time of day."
`README.md`	Basic project documentation	gpt-4o-mini	"Generate a short README file describing Python scripts that greet users based on the current time."

Table 1: Summary of AI-generated files and prompts used

Weekly Diary - All phases

You are required to submit a diary at the end of each project phase, indicating the individual contributions you made to the project and the report. This should be a summary of your activities for each week. You should follow the following tabular format.

Group Name		
Week	**Student Name**	**Activities**
W03	Jane Doh	Joined the meeting to discuss ideas and contributed to ideation
W03	John Doh	Joined the meeting to discuss ideas and contributed to ideation
W04	Jane Doh	Drafted project scope
W04	John Doh	Made the listed of planned activities
W05	...	...